AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 17 · STUDENT LESSON

Model Memory vs Retrieval

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 15: a source must be current and relevant to the claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of retrieval and indexing. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **retrieval** and **index**ing.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

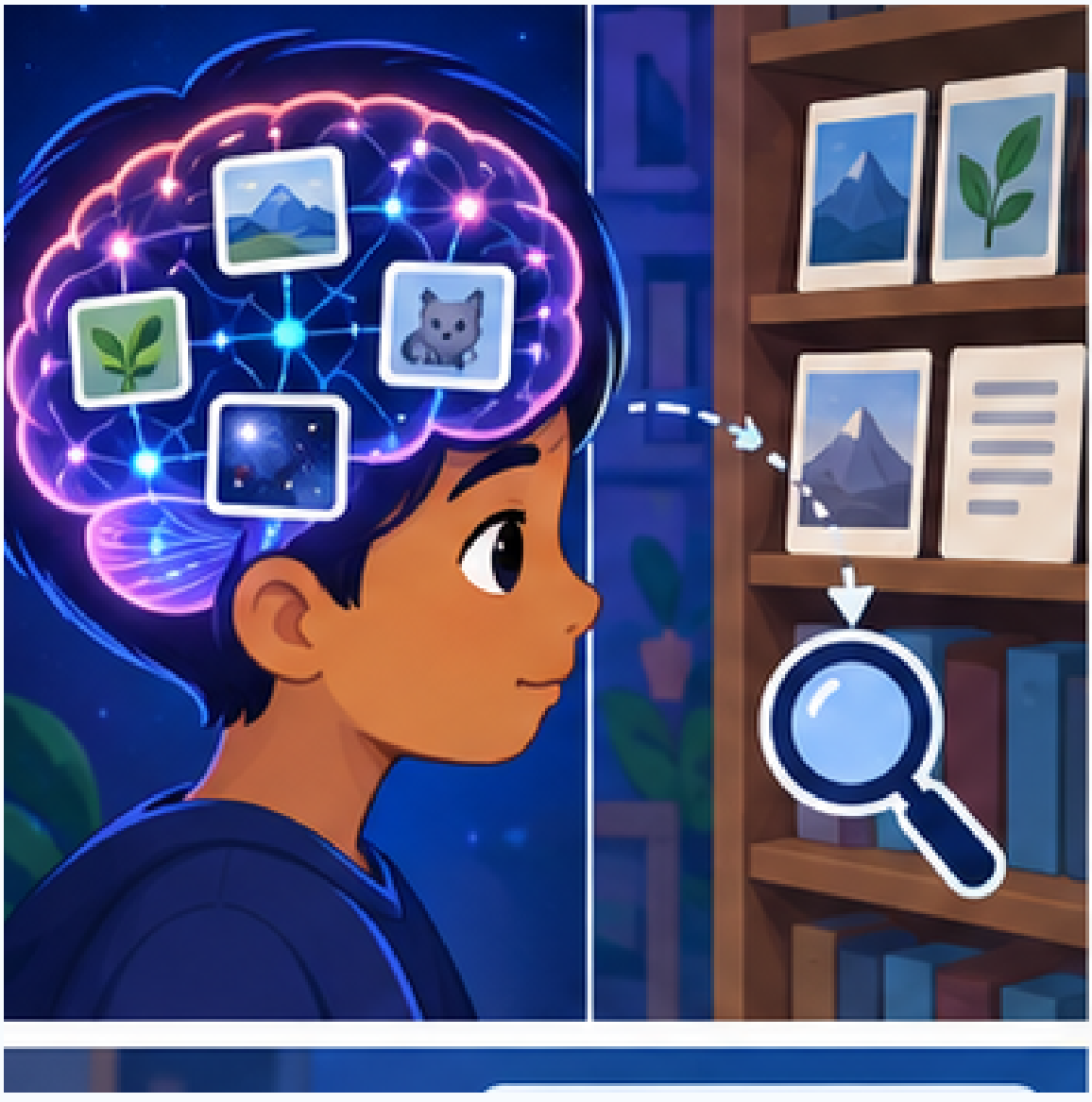
Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 17: Model Memory vs **Retrieval**

Learned **parameters** are not a searchable source library.



CREATOR MISSION

Decide when to answer, retrieve, calculate, or **abstain**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Speech pipelines and multimodal **evidence**. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can explain model memory vs **retrieval** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Model Memory vs Retrieval case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use model memory vs **retrieval** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Retrieval returns stored records selected for a query. An **index** is a searchable representation of a versioned **corpus**; generated wording is a later operation.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

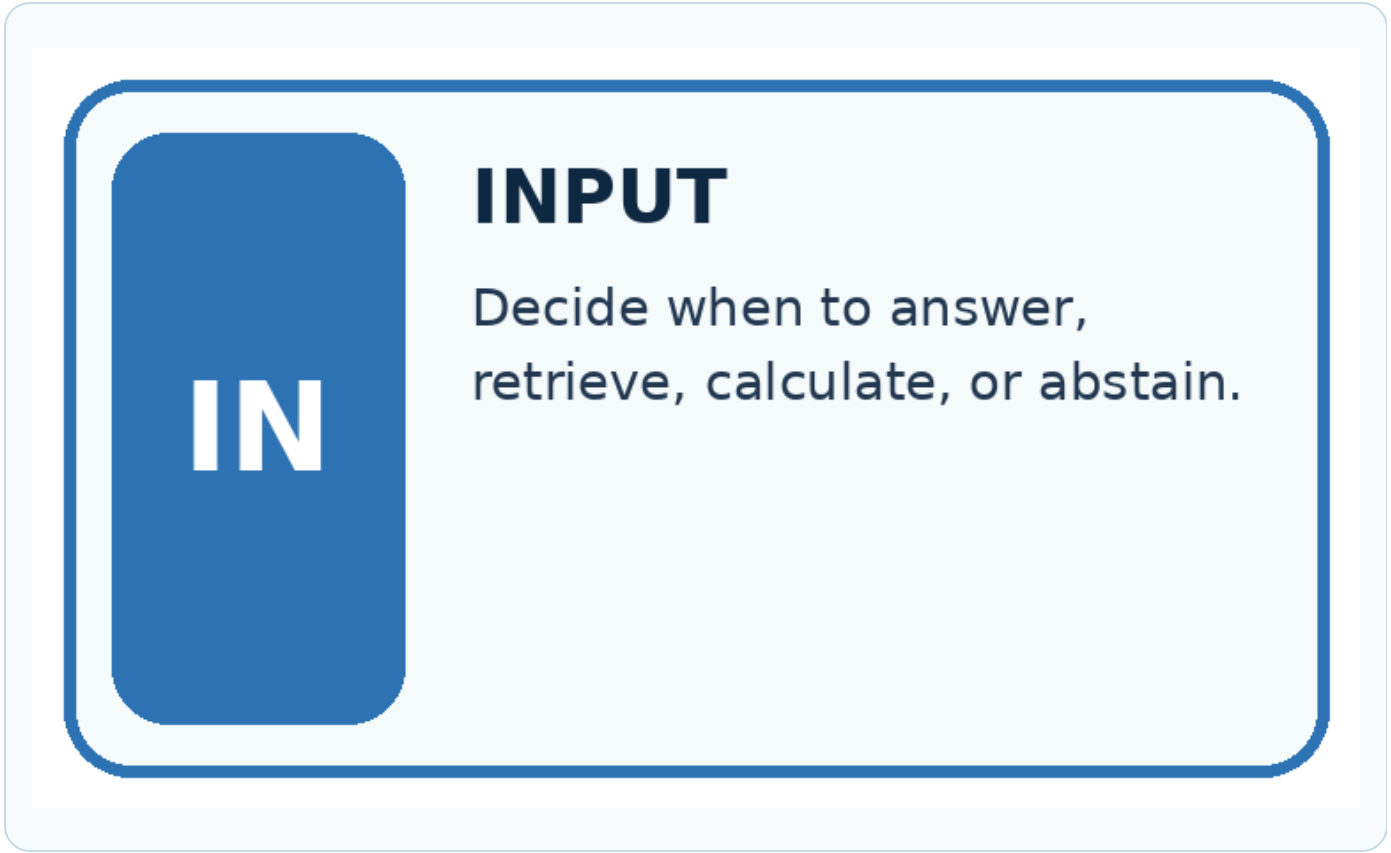
See how it works

> Follow the mechanism for Model Memory vs Retrieval. Connect each labelled input, deciding step and result.

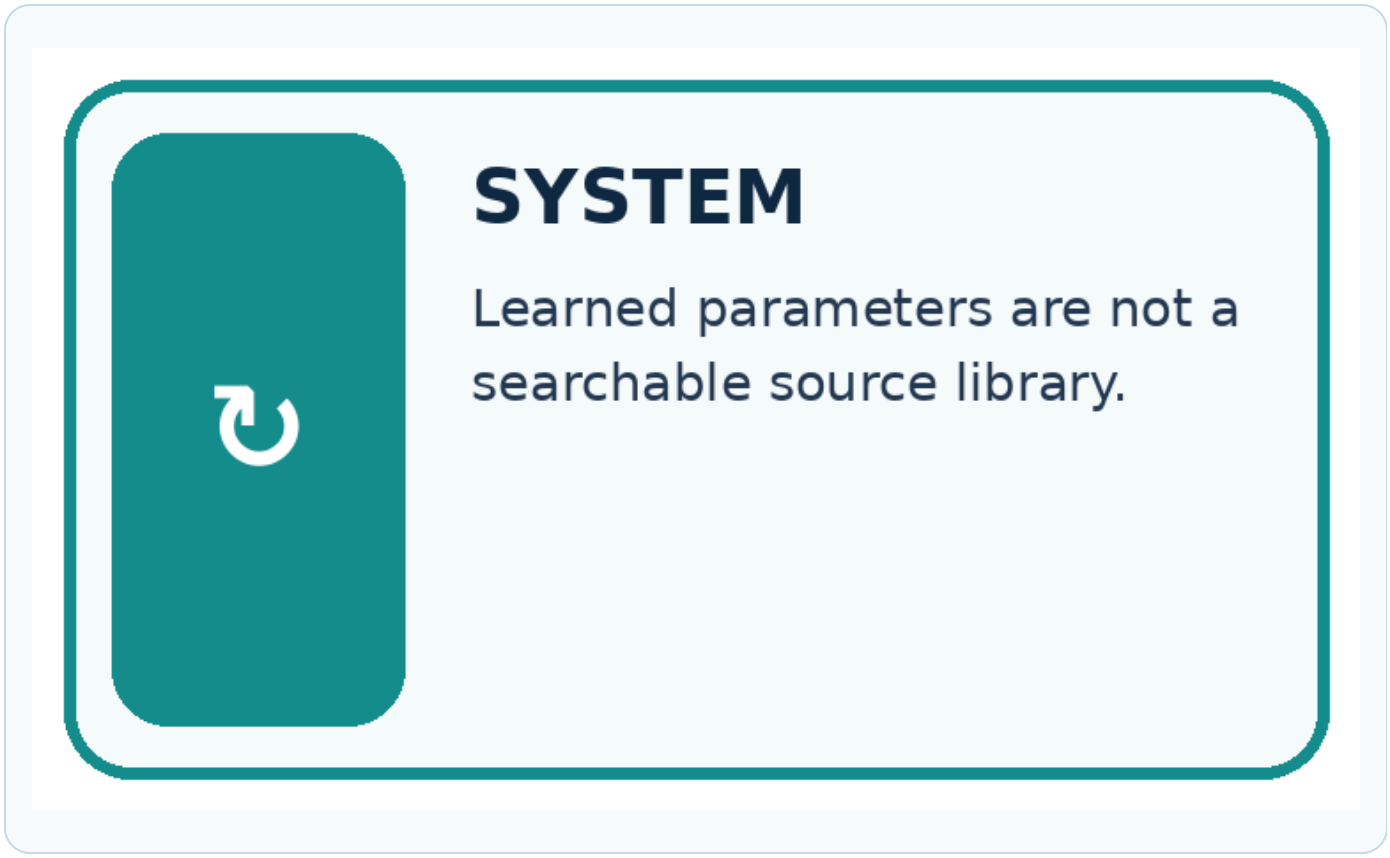
VISUAL MECHANISM

Input → Process → Output

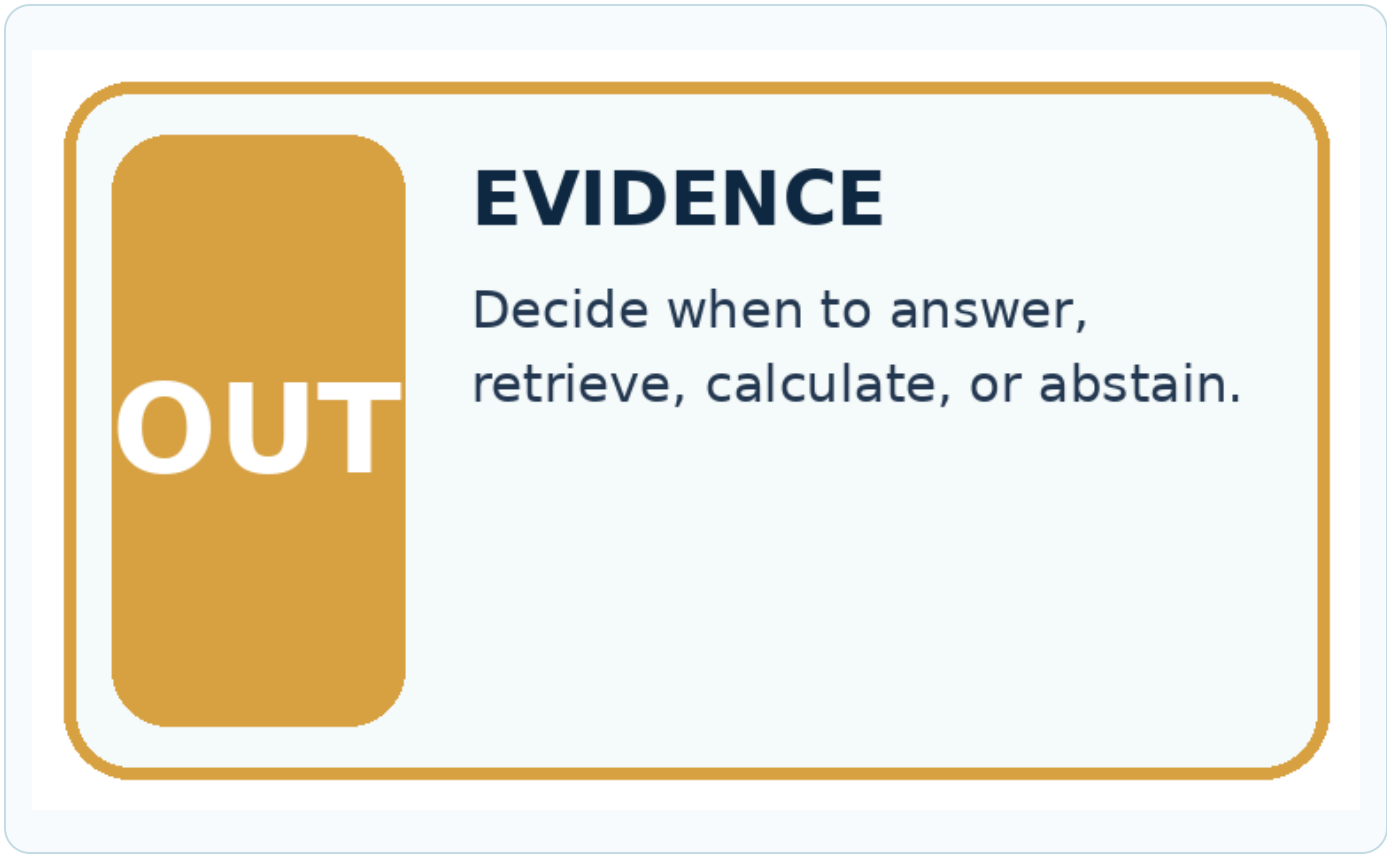
Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Learned parameters are not a searchable source library.

↓


OUT
EVIDENCE
Decide when to answer, retrieve, calculate, or abstain.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records. Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

Retrieval and indexing — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Index v1 contains notice N1 for room A. Corpus v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported?	Assign stable document IDs, versions and permitted access before indexing. Retrieve only authorised records. Preserve source references and define behaviour when a document is missing, outdated or access-restricted.	The stale index returns old evidence. It cannot establish current room approval; refresh against authorised corpus v2 or report unresolved freshness.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

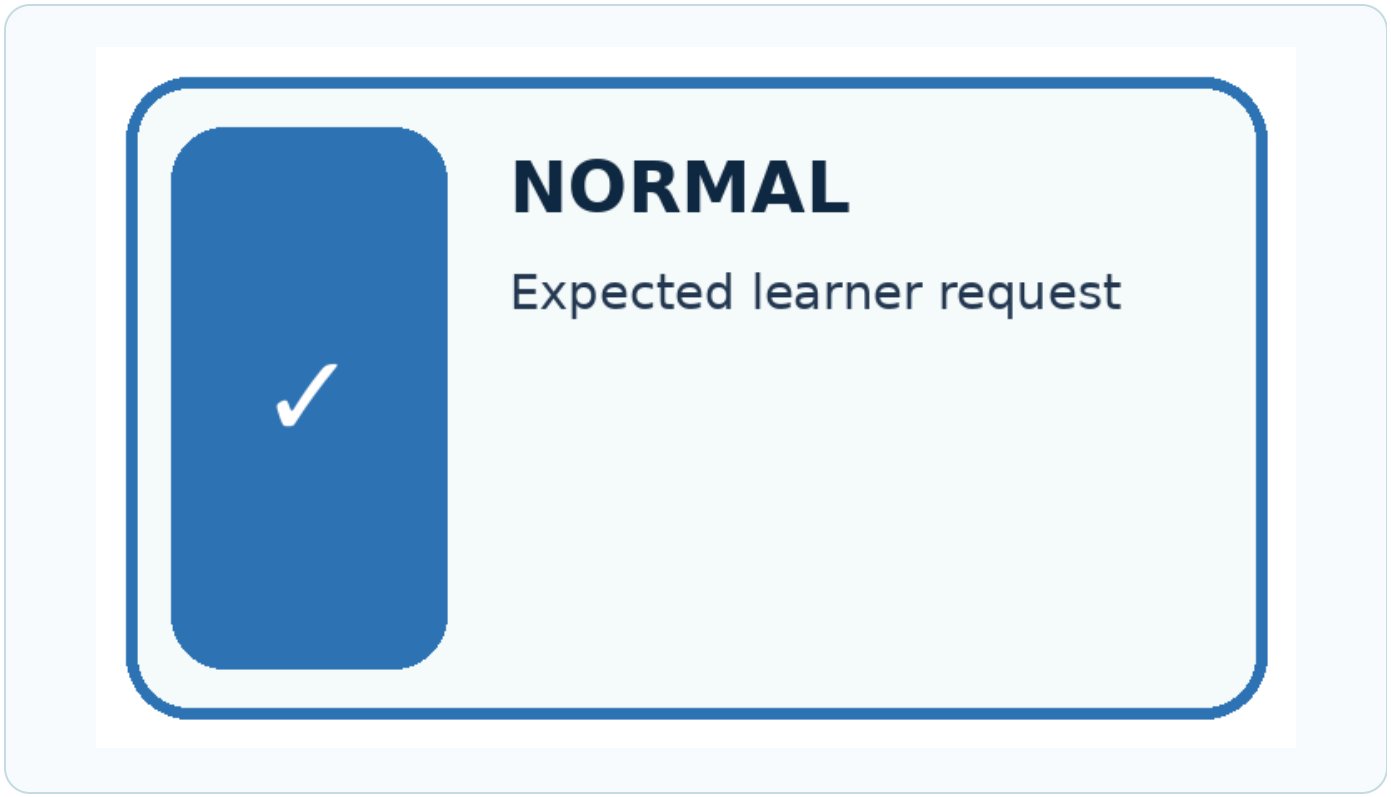
> Compare the supplied cases for Model Memory vs Retrieval. Identify the detail that changes the conclusion.

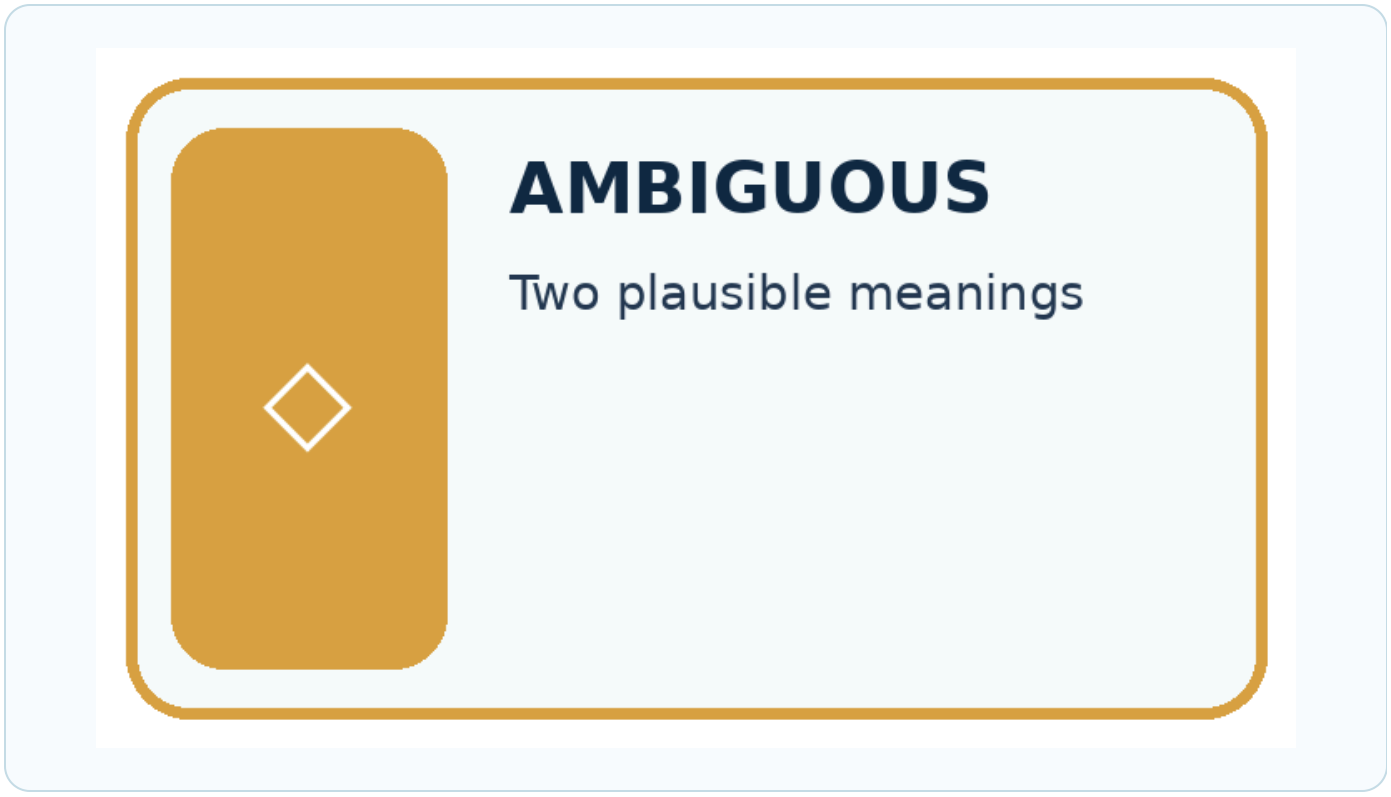
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: LIB-v = 2 is unavailable.

Predicted result

Prediction: Report that the current time cannot be verified.

Why the result follows

Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

STUDENT LESSON · 5 OF 18

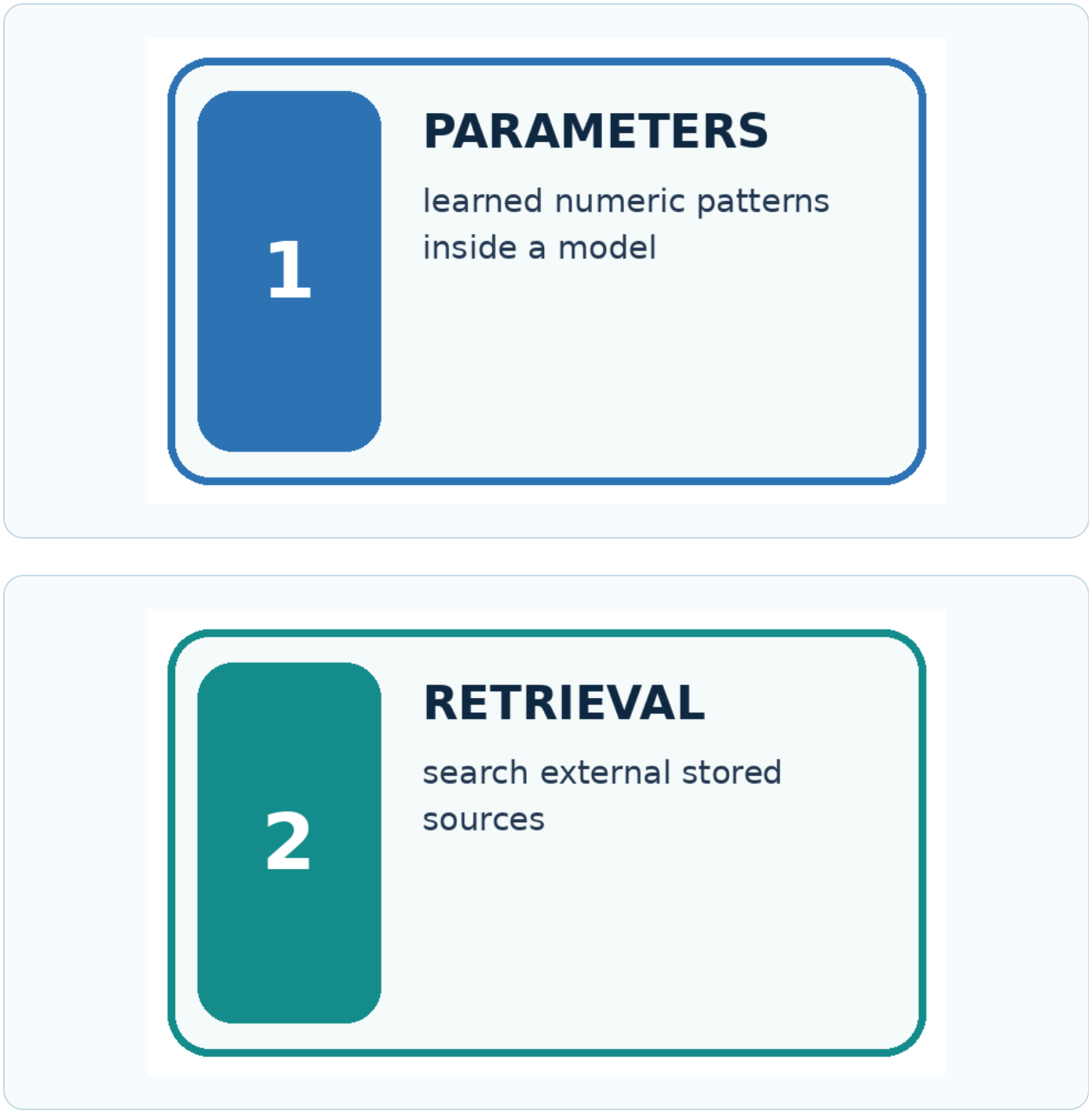
Words that unlock the idea

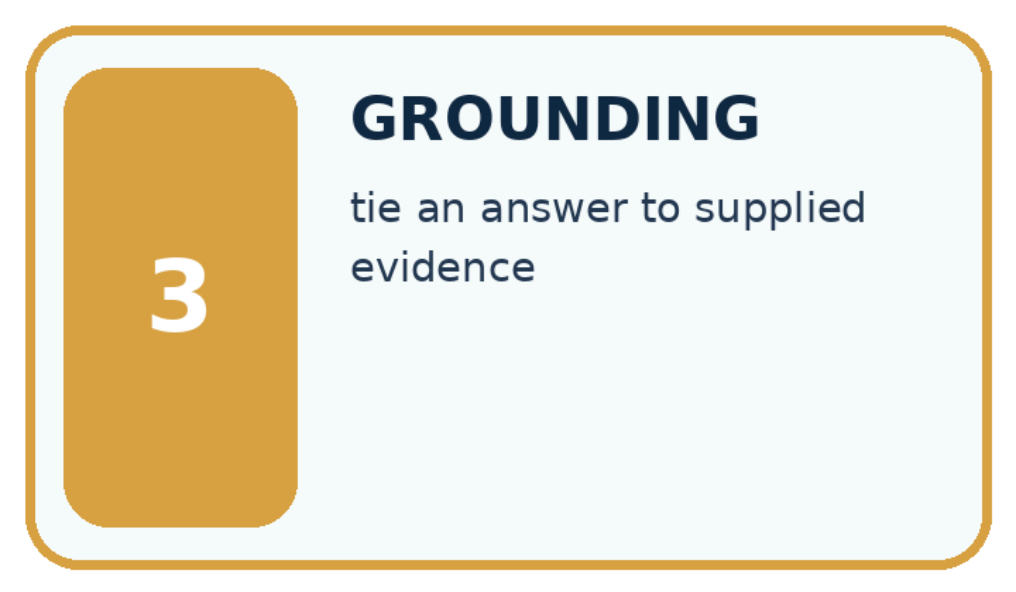
> Use the vocabulary of Model Memory vs Retrieval to explain a concrete decision, rather than only memorising definitions.

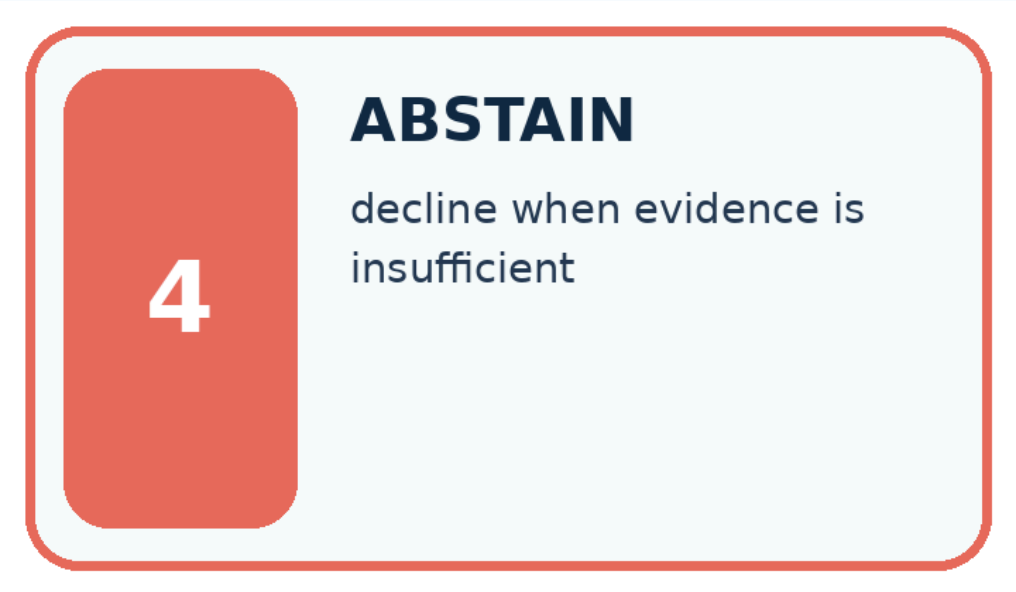
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
PARAMETERS
learned numeric patterns inside a model
2
RETRIEVAL
search external stored sources





Word	My own example	Not this / boundary
parameters	Values within a model adjusted during training; distinguish them from settings chosen by a developer.	
retrieval	Finding and returning existing information from a source or collection.	
grounding	Connecting an answer or action to specified evidence or observations.	

Word	My own example	Not this / boundary
abstain	Decline to make a prediction when the available evidence does not meet the decision rule.	

Words used in the closer explanation

Word	Meaning
Corpus	The collection of records available for a task
Index	A searchable representation of that collection
Freshness	Whether a record is current for the requested time

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Model Memory vs Retrieval, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 17 • WORKED EVIDENCE

1. Model memory may produce a plausible general explanation.
2. A current date or exact quotation requires retrieval.
3. Arithmetic should be calculated, not guessed from patterns.
4. When source or tool is unavailable, state the limit and abstain.

1. Model memory may produce a plausible general explanation.

2. A current date or exact quotation requires **retrieval**.

3. Arithmetic should be calculated, not guessed from patterns.

4. When source or tool is unavailable, state the limit and **abstain**.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Index v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported?

The stale **index** returns old **evidence**. It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

Worked explanation · Model Memory vs Retrieval

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Model Memory vs Retrieval

The library assistant remembers an old opening time, but a current notice can be retrieved.

The facts we will use

Supplied facts

Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00. The question asks the current Wednesday closing time.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/17

Model Memory vs Retrieval

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Old guess

A remembered-looking answer says 15: 00; no current notice is attached.



2. Find notice

Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00.



3. Ground answer

Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

CHANGE ONE THING

LIB-v = 2 is unavailable.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Old guess	A remembered-looking answer says 15: 00; no current notice is attached.
2. Find notice	Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00.
3. Ground answer	Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

Keep the stated limit

Read the labelled picture: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: LIB-v = 2 is unavailable. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Model Memory vs Retrieval claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A language model can inspect where every learned fact came from.
Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

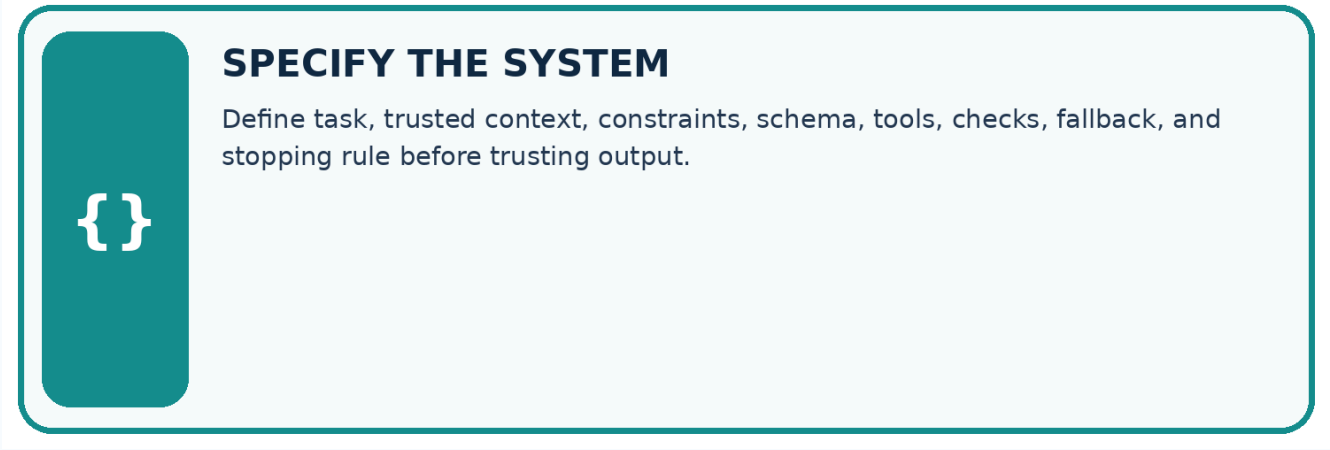
Practise together

> Practise Model Memory vs Retrieval with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, **retrieval**, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or **retrieval**_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or **abstain** RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

High **retrieval** similarity does not override permissions or make an obsolete source current.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Retrieval** and **index**ing". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Retrieval** and **index**ing. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

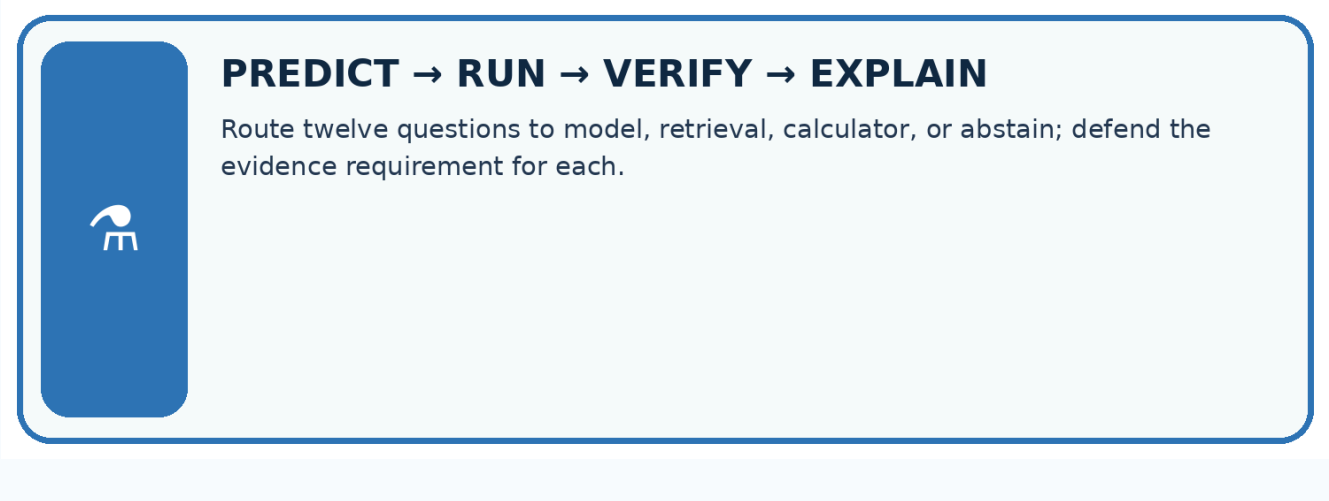
Run the investigation

> Investigate Model Memory vs Retrieval. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Route twelve questions to model, **retrieval**, calculator, or **abstain**; defend the **evidence** requirement for each.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

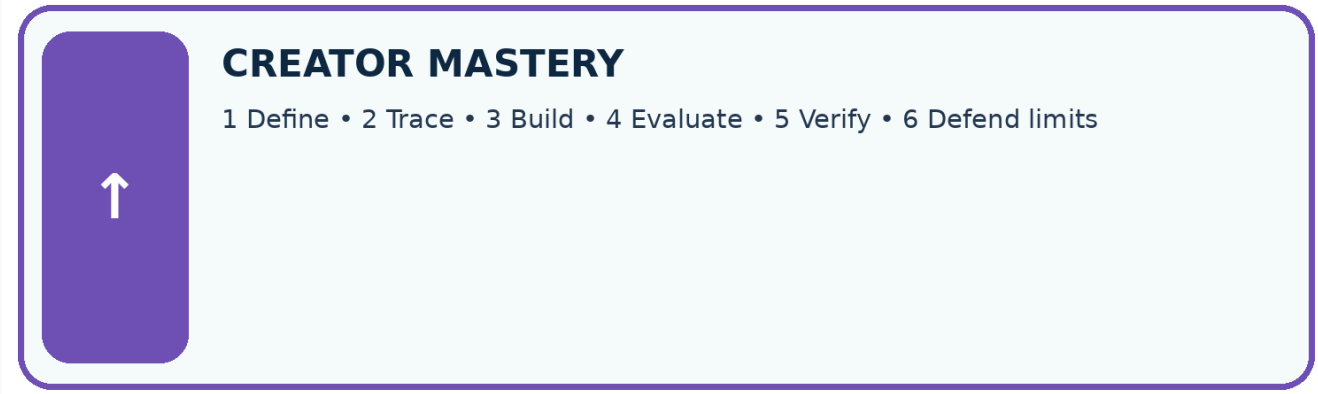
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Model Memory vs Retrieval. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

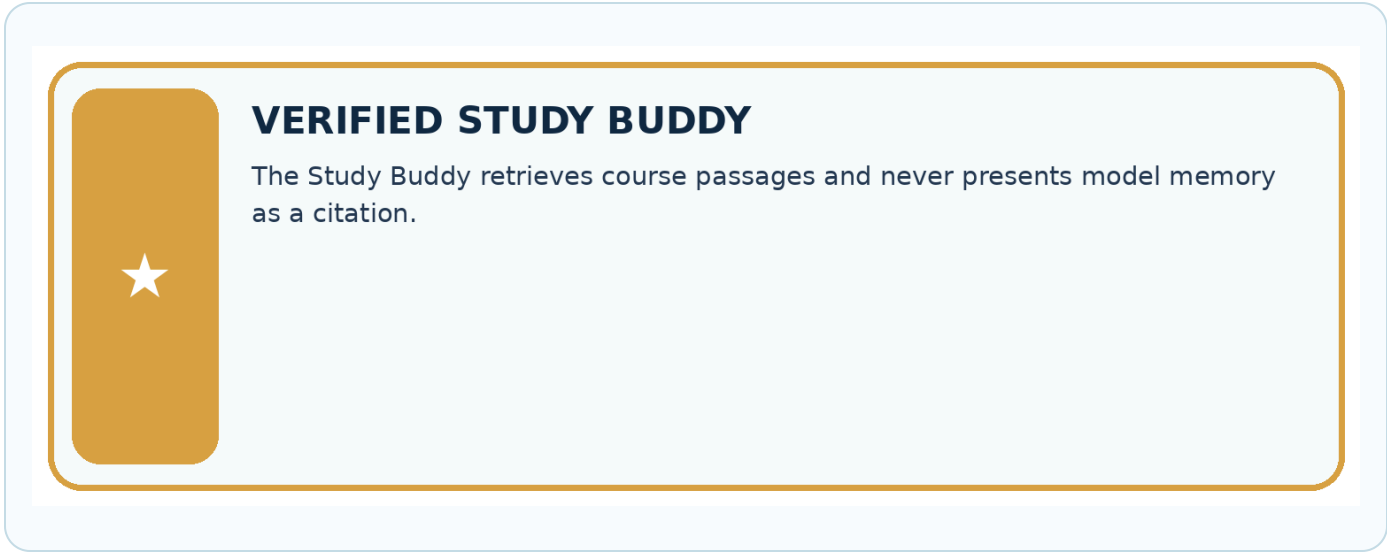
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT The Study Buddy retrieves course passages and never presents model memory as a citation.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/**retrieval** → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

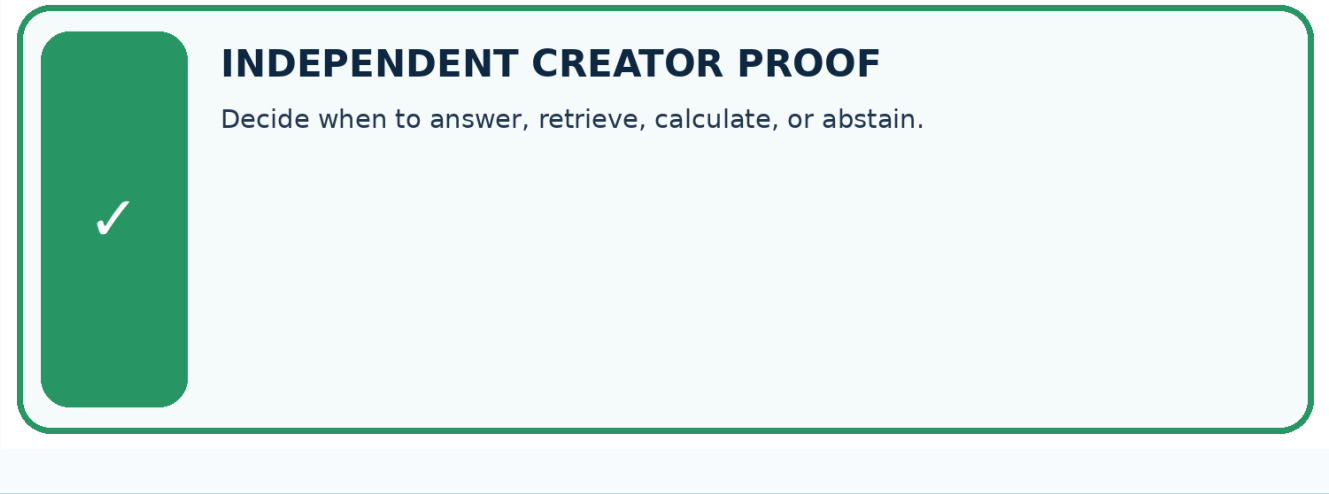
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Decide when to answer, retrieve, calculate, or **abstain**.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Model Memory vs Retrieval case for supported practice. Repeated examples do not become fresh assessments.

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

One assistant answers without search; another retrieves a handbook paragraph.

The question

Compare **evidence** trails.

Worked explanation

The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. **Retrieval** still does not guarantee correct use.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Retrieval creates an inspectable evidence trail

Evidence or stage	Value or result
No retrieval	Answer from learned patterns
Retrieval	Inspect a handbook paragraph
Verification	Check the paragraph supports the answer

One assistant answers without search; another retrieves a handbook paragraph.

Compare **evidence** trails.

The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. **Retrieval** still does not guarantee correct use.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Corpus has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A model recalls last year's opening hours; a retrieved current notice gives different hours. How should the answer use this **evidence**?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

The Study Buddy retrieves course passages and never presents model memory as a citation.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A model states the library opens at 8 from learned patterns. A supplied current notice says 9. Explain the stronger answer and one remaining check.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Model Memory vs Retrieval, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: The library assistant remembers an old opening time, but a current notice can be retrieved.

By the end, I can	How I will show it
Explain the mechanism and terminology of retrieval and indexing	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Model Memory vs Retrieval without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **retrieval** and **index**ing

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

The library assistant remembers an old opening time, but a current notice can be retrieved.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

1. Old guess

2. Find notice

3. Ground answer

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Ground answer Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Read the picture in order.

Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: LIB-v = 2 is unavailable. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Report that the current time cannot be verified.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report that the current time cannot be verified. The unsupported old guess does not become current **evidence** when **retrieval** fails.

Repair a missing building block

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Explain the **mechanism** and terminology of image generation and provenance. Short practice: Which checks pass after that single change, and what still needs repair?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

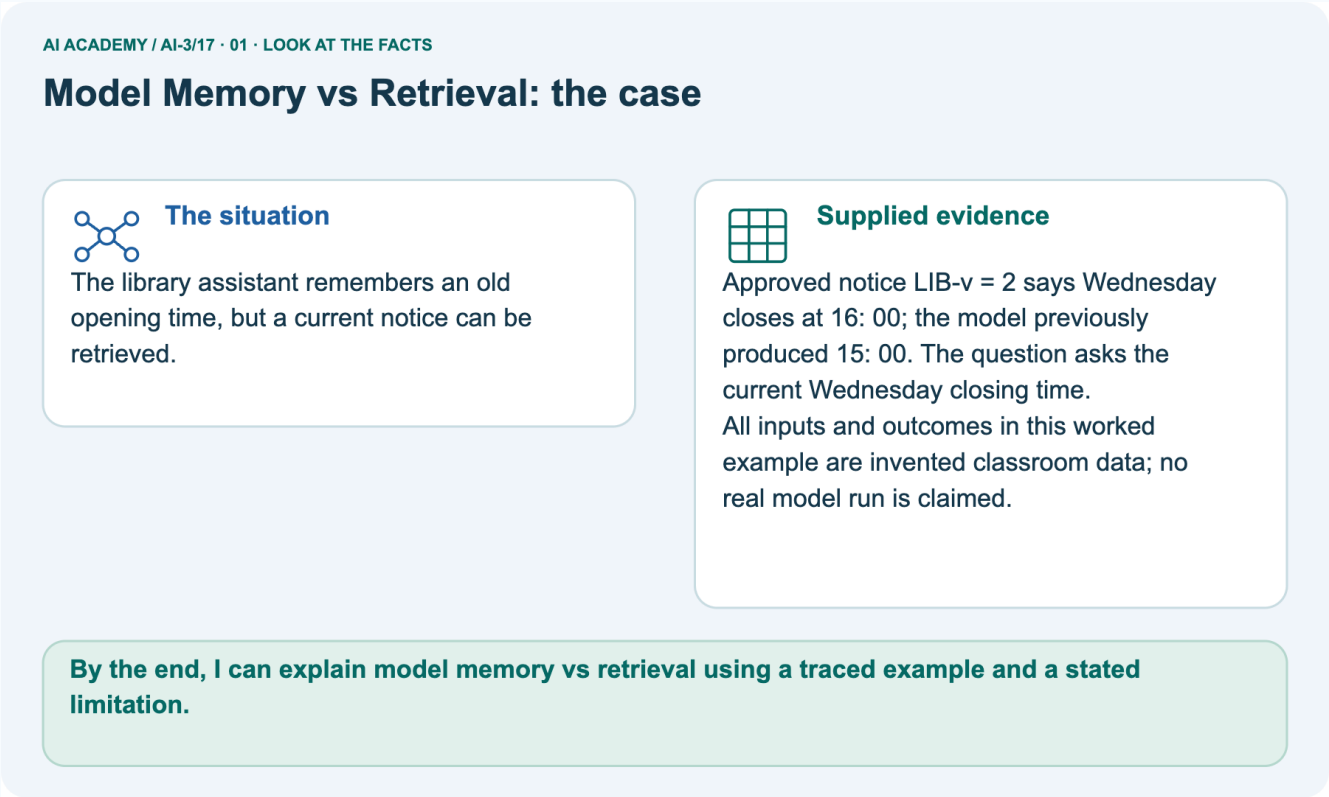
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Model Memory vs Retrieval graphic using its labels, deciding evidence and relevant limitation.

01. Model Memory vs Retrieval: the case



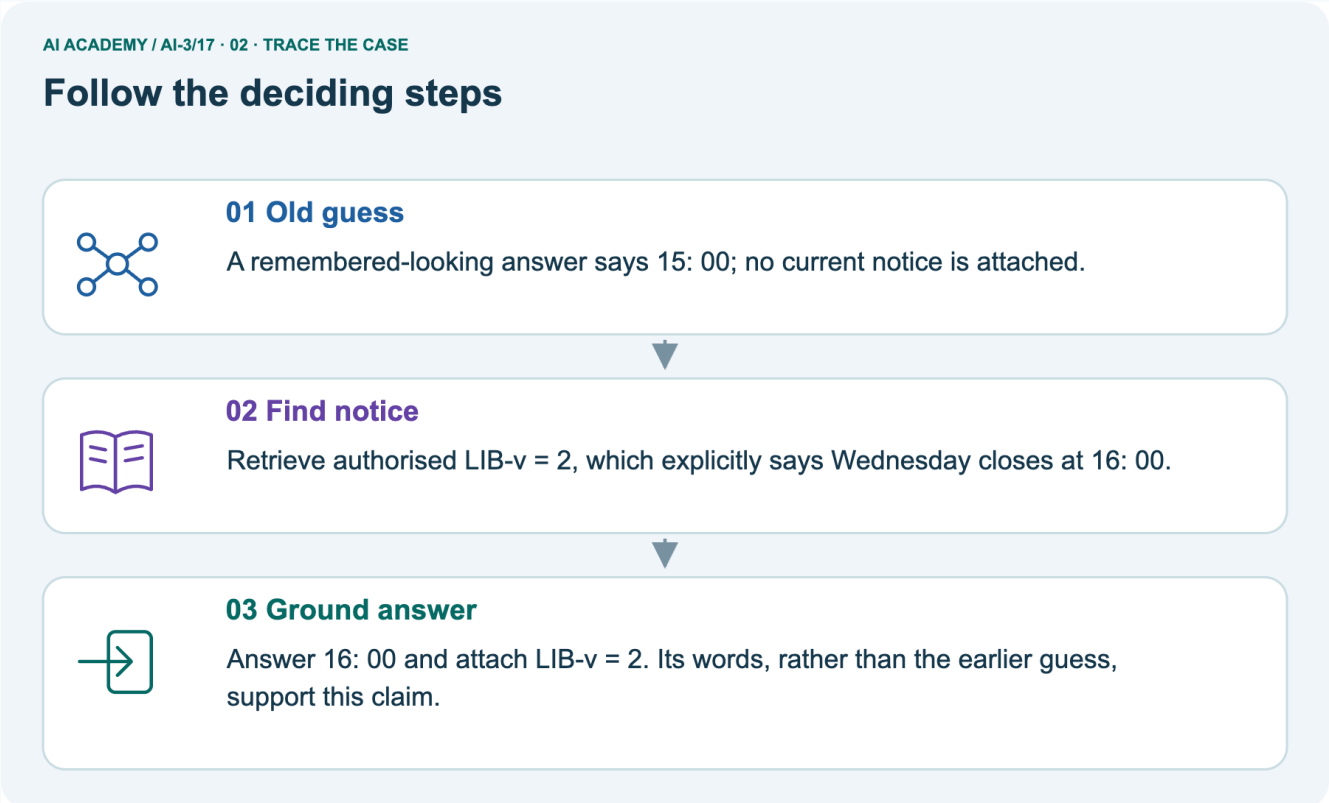
Follow the labelled picture with the matching explanation. Open the vector diagram

The library assistant remembers an old opening time, but a current notice can be retrieved. Supplied facts: Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00.

The question asks the current Wednesday closing time. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



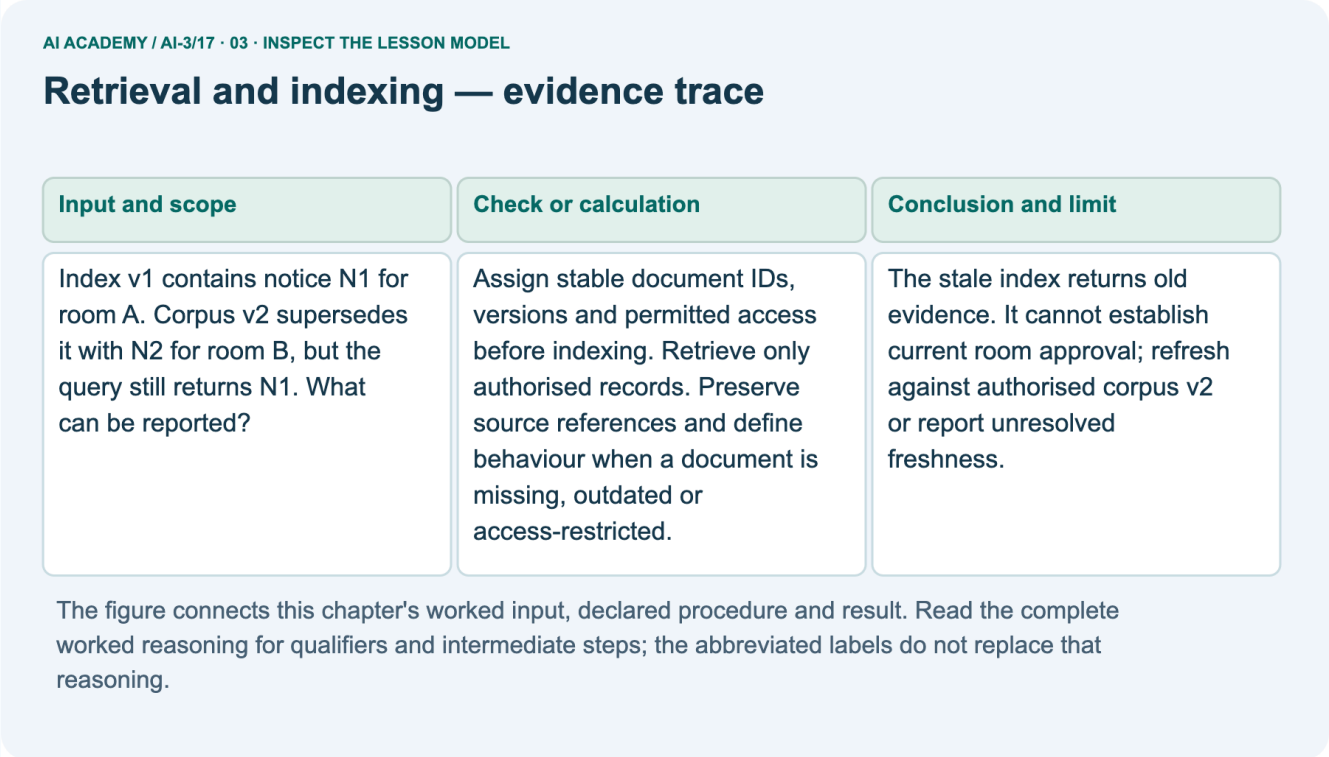
Follow the labelled picture with the matching explanation. Open the vector diagram

Old guess: A remembered-looking answer says 15: 00; no current notice is attached. → Find notice: Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00. → Ground answer: Answer 16: 00 and attach LIB-v = 2.

Its words, rather than the earlier guess, support this claim.

Point to each numbered stage and explain which recorded **input** it uses.

03. Retrieval and indexing — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieval and **index**ing — **evidence** trace. **Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1.

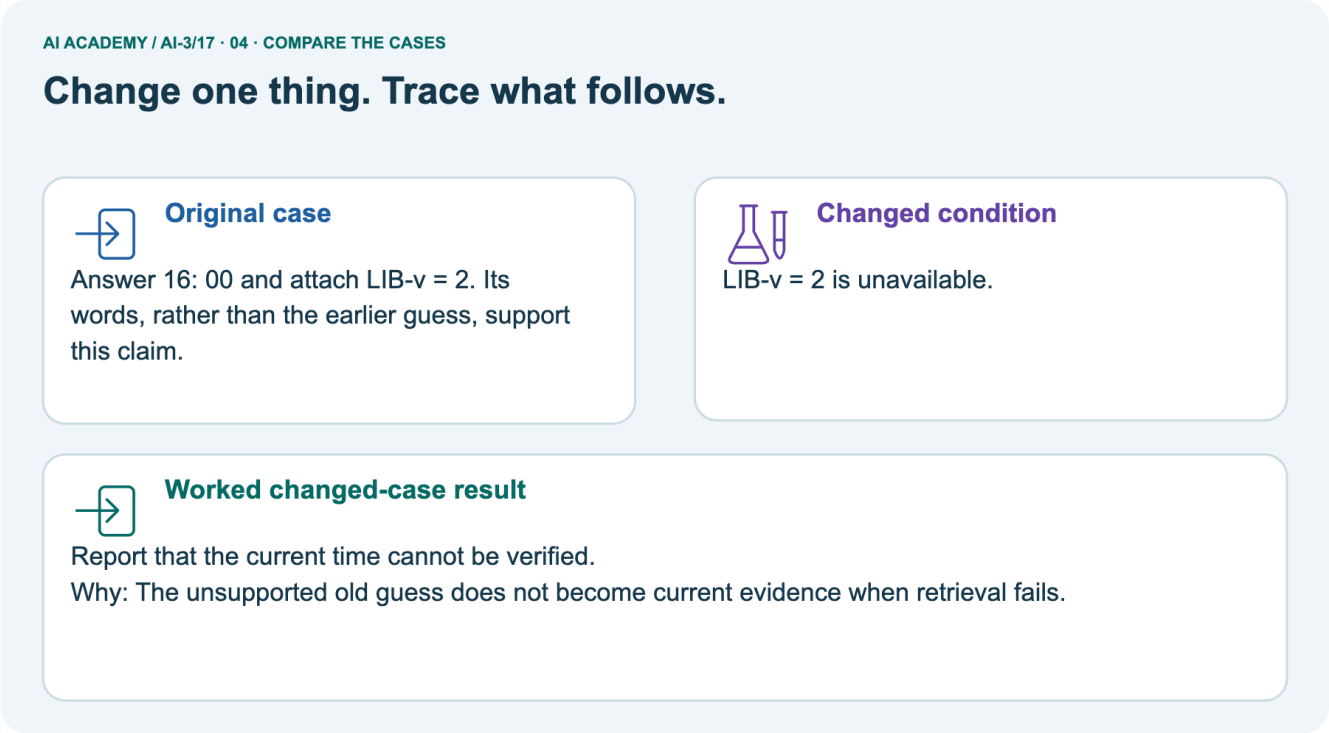
What can be reported? | Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records.

Preserve source references and define behaviour when a document is missing, outdated or access-restricted. | The stale **index** returns old **evidence**.

It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



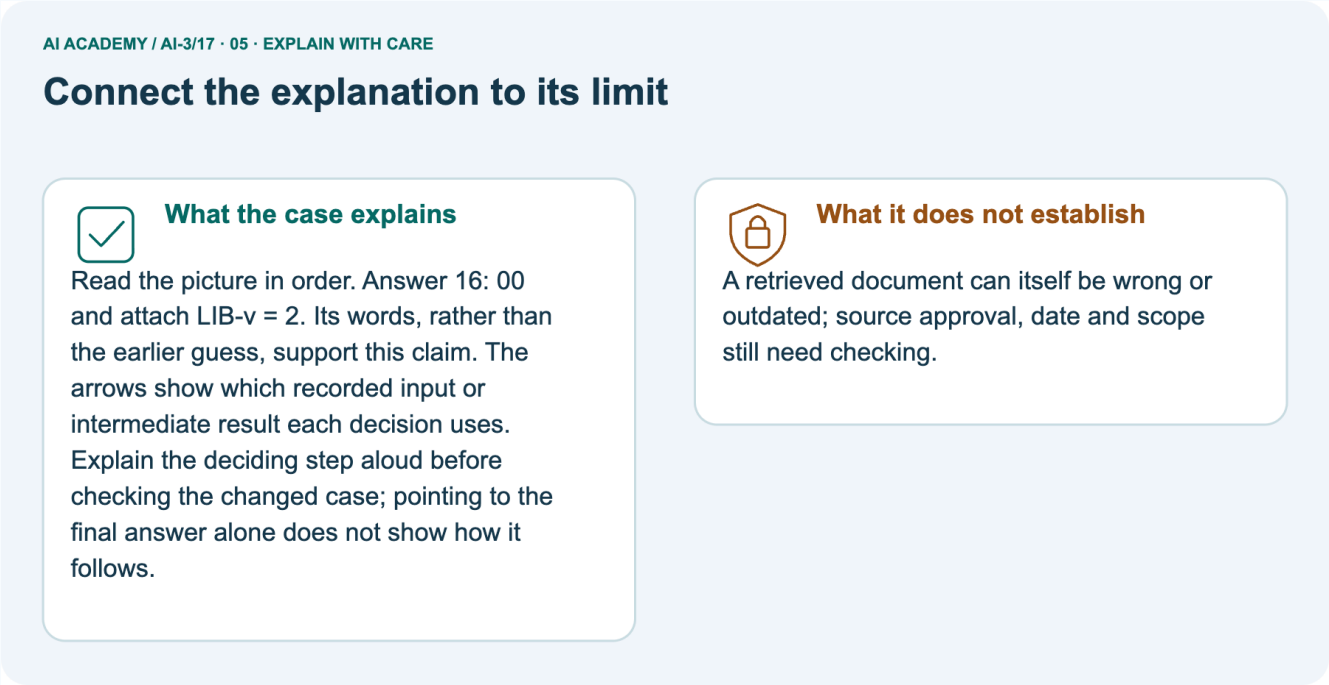
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Changed condition: LIB-v = 2 is unavailable. Expected result: Report that the current time cannot be verified.

Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

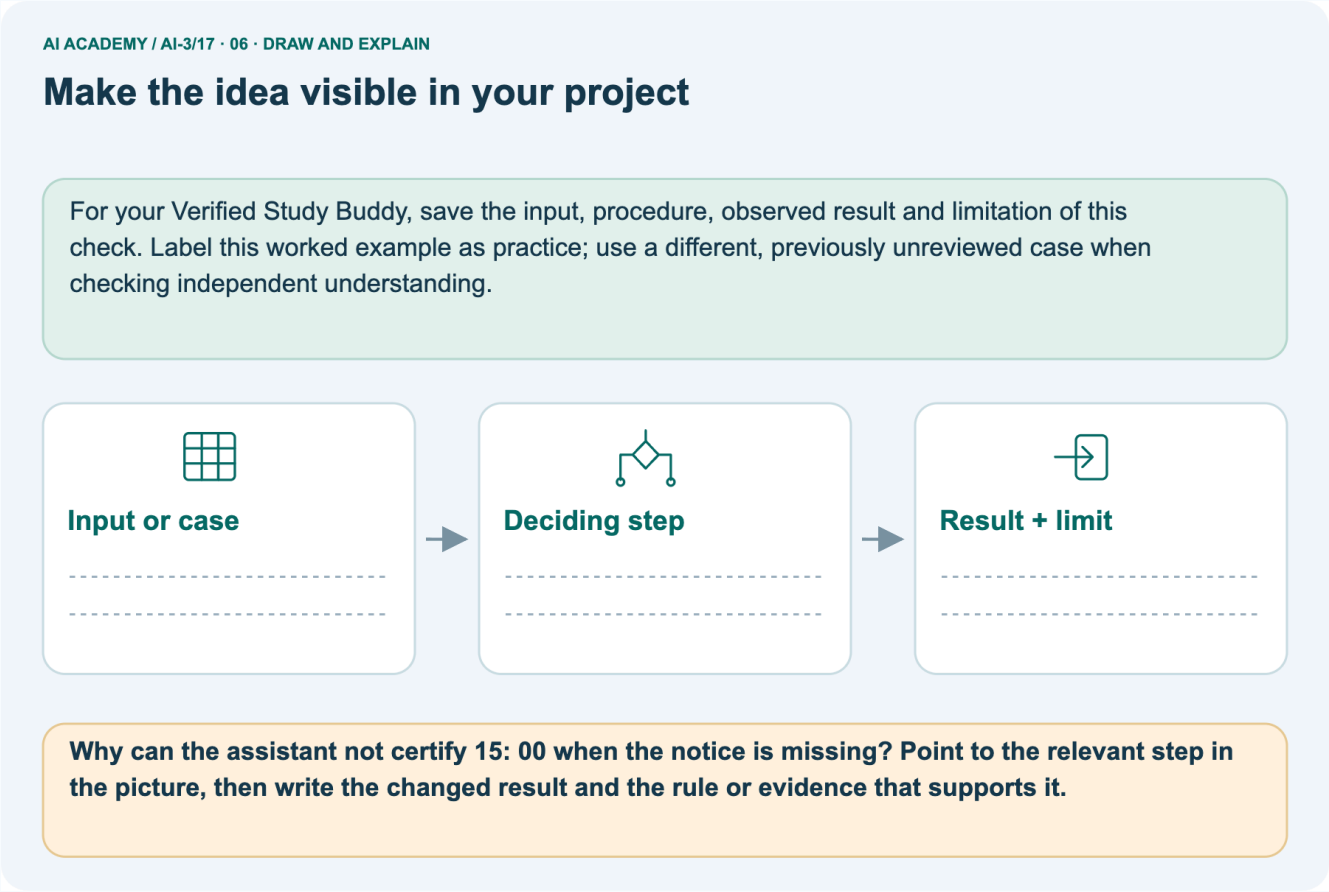
Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.